جامعة بيرزيت

BIRZEIT UNIVERSITY

Software Testing And Quality Assurance

Comp438

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Introduction

This report includes the comprehensive quality assurance process applied to the YouTube website, selected as the system under test due to its extensive features, functionality, and widespread usage

The purpose of this project is to ensure the quality, reliability, and performance of the YouTube website through systematic testing

The testing process covers requirements analysis, test planning, test case development, execution, automation, and performance testing, the project aims to enhance user experience and verify the system capability to handle diverse scenarios effectively

Additionally, this report highlights the tools, techniques, and methodologies utilized throughout the testing process and results

Requirements Analysis

The YouTube website is a versatile system offering a variety of functionalities, analyzing both functional and nonfunctional requirements is essential to ensure good operation and comprehensive testing

Functional requirements:

- **R1: Video playback**
  - Users should be able to play, pause, rewind, and fast forward videos
  - Support different video resolutions
- **R2: Search functionality**
  - Users should be able to search for videos using keywords
  - Provide autocomplete suggestions while typing in the search bar
- **R3: Account management**
  - Users should be able to sign up, log in, and manage their profiles
  - Enable features like viewing watch history and managing playlists
- **R4: Video upload**
  - Users can upload videos in supported formats with specific size and duration limits
  - Allow users to add titles, descriptions, tags
- **R5: Comments**
  - Enable replies to comments and like, dislike comments
- **R6: Interaction**
  - Allow users to like, dislike videos, and share videos
- **R7: Recommendations**
  - Suggest videos based on the users viewing history and preferences
- **R8: Live streaming**
  - Users should be able to broadcast live videos
  - Users should be able to make real time chat during live streaming
- **R9: Subscription**
  - Users can subscribe to channels to follow content updates
- **R10: Notifications**
  - Users should be notified of new videos or live streams from subscribed channels

- **R11: Captions and transcriptions**
  - Provide auto-generated captions in multiple languages
  - Users should edit the caption of their own videos
- **R12: Download in offline videos**
  - Users should be able to download videos for offline viewing
- **R13: Content moderation**
  - Admins should be able to ban or suspend user accounts violating terms of service
  - Provide tools to flag inappropriate content
  - Review and manage reported content
- **R14: Analytics**
  - Provide content creators with video performance metrics like views, likes, engagement

Non functional requirements:

- **R1: Scalability**
  - The system shall be able to handle millions of users worldwide without noticeable decrease in the performance
- **R2: Caching**
  - frequently accessed videos and search should be cached to reduce loading time
- **R3: Fraud detection**
  - The system should contain algorithms to detect fake accounts and bot activity
  - The system should prevent ad click fraud
- **R4: Secure payment**
  - The system should provide a secure payment for premium features
- **R5: Well documented and standardized database**
  - The system should be prepared for periodic updates
  - The system should support the routine maintenance without disturbing user experience
- **R6: Supports diversity**
  - The system should support various devices, operating systems, and browsers
  - The system should be compatible with multiple video and audio codecs

Test Plan

This section contains a structured test plan for the YouTube website

Introduction:

- **Objective:** the purpose of this test plan is to verify the functionalities and performance of the YouTube website, ensuring it meets functional and non-functional requirements for users

Scope:

- **Scope:** the testing will cover the functional, performance, and usability aspects of the YouTube website, including the login, commenting, video playing, and searching features, this excludes advanced security testing and all non-user-facing features, such as backend analytics

Test objectives:

- **Functional testing objectives:** verify the valid login functionality, valid comment submission, login with incorrect password, entering mismatched passwords into the reset form, search functionality verification, video playback validation, invalid file type upload, empty comment submission, notifications functionality, share video, verify notification access while offline, view history with invalid conditions
- **Performance testing objectives:** ensure that YouTube can handle very concurrent users interacting with the system
- **Usability testing objectives:** evaluate the ease of use of the search bar, video player, and comment sections

Test strategy:

- **Approach:** the agile testing approach will be used with manual and automated testing
- **Testing levels:** validate the overall functionality of the YouTube website
- **Techniques:** the black-box testing for user scenario by Selenium, load testing using JMeter for concurrent users

Test schedule:

- **Functional testing:** starts on [04/01/2025] and ends on [08/01/2025]
- **Performance testing:** starts on [10/01/2025] and ends on [12/01/2025]
- **Usability testing:** starts on [04/01/2025] and ends on [08/01/2025]

Test resources:

- **Hardware:** we will test the YouTube website on different types of devices to make sure it works well for everyone
- **Software:** the Selenium for automated testing, JMeter for performance testing
- **Personnel:** four team members, each assigned specific testing tasks

Test deliverables:

- **Document structure:** test plan, test cases, bug reports, test automation, performance testing

Risks and assumptions:

- **Assumptions:** the test environment will be fully set up and stable before testing begins
- **Risks:** the test environment might not match the production environment exactly, leading to inaccurate results

Dependencies:

- **External dependencies:** the testing schedule is dependent on the completion of development tasks

Exit criteria:

- **Exit criteria testing will be considered complete when:** no critical issues are found in the final test, all functional, performance, security, and usability tests have been executed

Approvals:

- **The test plan requires approval from:** project manager, QA manager

Appendices:

- **Glossary:** the test case is a specific set of inputs, execution conditions, and expected results developed to verify a particular feature
- **References:** https://support.google.com/youtube#topic=9257498

Test Casas & Test Execution

The section contains all the test cases, with positive test cases and negative test cases, the test cases were developed using well black box testing, to manage and document these test cases efficiently, using Jira, a specialized testing and project management tool

Positive test case:

TC: Search functionality verification (link in Jira)

Projects / MyProject1 / Add epic / MP1-1

TC05: Search functionality verification

+ Add Apps

Description
Verify that the search functionality on youtube returns right results when a user enters valid keywords

Pre conditions
Stable internet connection is available

The user is logged in

The youtube homepage is accessible

Test steps
Navigate to the youtube homepage

Locate the search bar at the top of the page

Enter the keyword "technology"

Click on the search button or press enter

Test data
Keyword "technology"

Expected result
The search results page should display a list of videos related to the keyword "technology"

The titles and descriptions of the results should include or relate to the keyword

Actual result
Display a list of videos related to the keyword "technology"

Status
Pass

Priority
High

HM Add a comment...

Analyze the results:

Date: 4/1/2025

Expected result: the search results page should display a list of videos related to the keyword “technology” and the titles and descriptions of the results should include or relate to the keyword

Actual result: display a list of videos related to the keyword “technology”

Conclusion: it worked as expected, displaying relevant videos for the keyword "technology"

Detect bugs report:

No bugs detected

TC: Video playback validation (link in Jira)

Projects / MyProject1 / Add epic / MP1-4

TC06: Video playback validation

+ Add Apps

Description
Verify that videos on youtube play without interruptions when selected

Pre conditions
Stable internet connection is available

User is logged in

User is on the video page

Test steps
Navigate to the youtube home page

Click on any video thumbnail to open the video

Verify that the video starts playing automatically or after clicking the play button

Observe the playback for at least 20 seconds

Test data
Any video so select any trending video or a video from search results

Expected result
The video plays smoothly without interruptions

Actual result
The video plays without interruptions

Status
Pass

Priority
High

HM Add a comment...

Analyze the results:

Date: 4/1/2025

Expected result: the video plays smoothly without interruptions

Actual result: the video plays without interruptions

Conclusion: it worked as expected, the video played smoothly and without any interruptions

Detect bugs report:

No bugs detected

Negative test case:

TC: Invalid file type upload (link in Jira)

Projects / MyProject1 / Add epic / MP1-6

TC07: Invalid file type upload

+ Add Apps

Description
Verify that the youtube prevents uploading files that are not supported

Pre conditions
Stable internet connection is available

User is logged in

User is on the video upload page

Test steps
Navigate to the video upload page

Click on the choose file

Select an invalid file type

Attempt to upload the selected file

Test data
Invalid file "test.txt"

Expected result
It should display an error message stating that the file type is not supported and prevent the upload

Actual result
It display invalid file format

Status
Pass

Priority
Medium

HM Add a comment...

Analyze the results:

Date: 4/1/2025

Expected result: it should display an error message stating that the file type is not supported and prevent the upload

Actual result: it displays invalid file format

Conclusion: the expected behaviour was achieved, so the error message correctly prevents the upload

Detect bugs report:

No bugs detected

TC: Empty comment submission (link in Jira)

Projects / MyProject1 / Add epic / MP1-7

TC08: Empty comment submission

+ Add Apps

Description
Verify that an empty comment can not be submitted

Pre conditions
Stable internet connection is available

User is logged in

A video page is open with the comment section visible

Test steps
Select a video

Navigate to the comment section of the video page

Leave the comment text box empty

Click the comment button

Test data
No data is entered into the comment text box

Expected result
The comment should not be submitted and an error message

Actual result
The comment button is disabled and can not be clicked

Status
Pass

Priority
Medium

HM Add a comment...

Analyze the results:

Date: 4/1/2025

Expected result: the comment should not be submitted and an error message

Actual result: the comment button is disabled and cannot be clicked

Conclusion: it worked as expected by disabling the comment button, preventing the empty comment from being submitted

Detect bugs report:

Bug description: no error message displayed, although the button is disabled

Expected result: the error message like comment cannot be empty, should appear along with the disabled button

Actual result: the comment button is disabled, but no error message is shown to the user

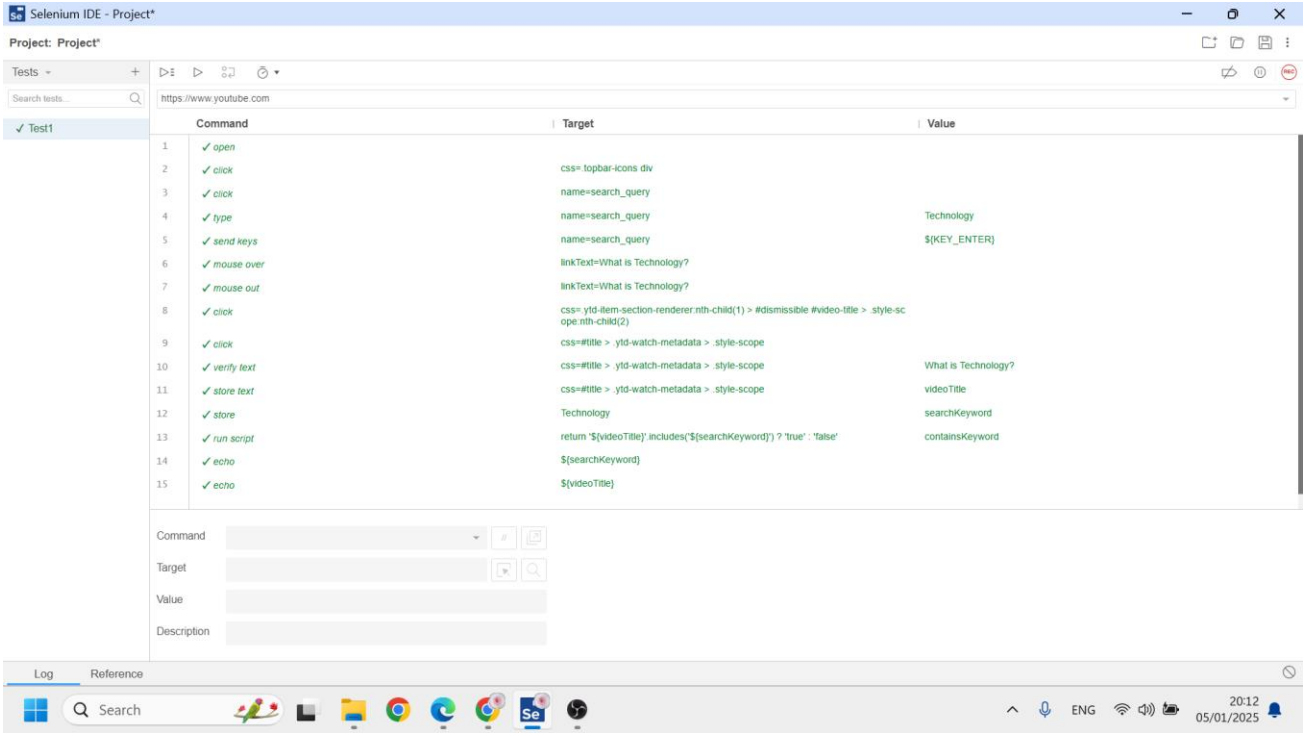
Table1: traceability matrix

Requirement	Requirement description	Priority	Test case	Test result
R3: Account management	Handles user registration, login, and profile updates	High	TC01: Valid login functionality	Pass
R5: Comment	Allows users to find content using keywords	Medium	TC02: Valid comment submission	Pass
R3: Account management	Handles user registration, login, and profile updates	High	TC03: Login with incorrect password	Pass
R3: Account management	Handles user registration, login, and profile updates	High	TC04: Entering mismatched passwords in the reset form	Pass
R2: Search functionality	Allows users to find content using keywords	High	TC05: Search functionality verification	Pass
R1: Video playback	Enables viewing videos with playback controls	High	TC06: Video playback validation	Pass
R4: Video upload	Facilitates uploading and storing videos	Medium	TC07: Invalid file type upload	Pass
R5: Comment	Supports user engagement through comments, likes, and replies	Medium	TC08: Empty comment submission	Pass
R10: Notification	Users should be notified of new videos or live streams from subscribed channels	High	TC09: Notifications functionality	Pass
R6: Interaction	Allow users to like, dislike videos, and share videos	High	TC10: Share video	Pass
R10: Notification	Users should be notified of new videos or live streams from subscribed channels	Medium	TC11: Verify notification access while offline	Pass
R3: Account management	Enable features like viewing watch history and managing playlists	Low	TC12: View history with invalid conditions	Pass

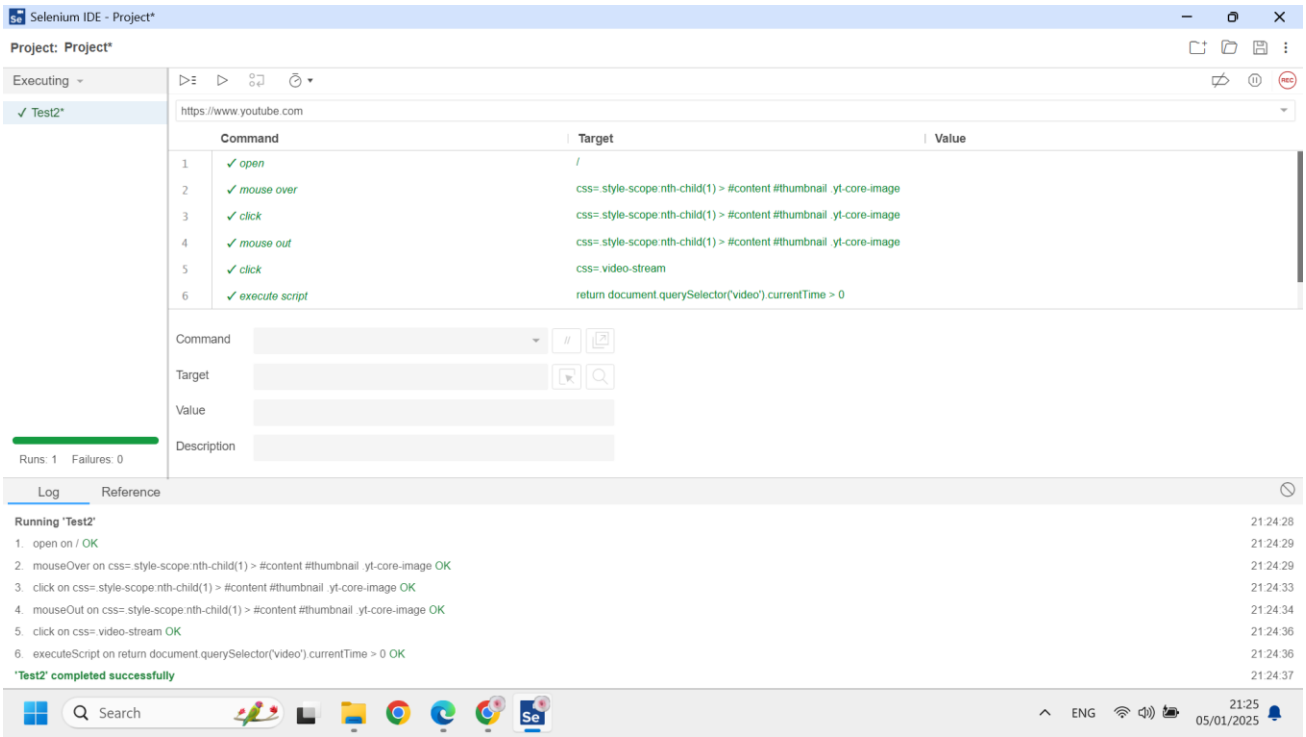
Test Automation

This section contains automated test cases using Selenium, ensuring efficient and reliable testing processes

TC: Search functionality verification



TC: Video playback validation



Performance & Load Testing

This section contains the performance and load testing for the YouTube website, using JMeter to evaluate the system performance under varying levels of user activity, the goal is to simulate real world scenarios

Scenario1:

Load testing: measure the YouTube performance to **the main page** under expected loads to ensure it can handle the anticipated user base without degrading performance

Thread properties: number of users: 300, ramp-up period (in sec): 2, loop count: 1

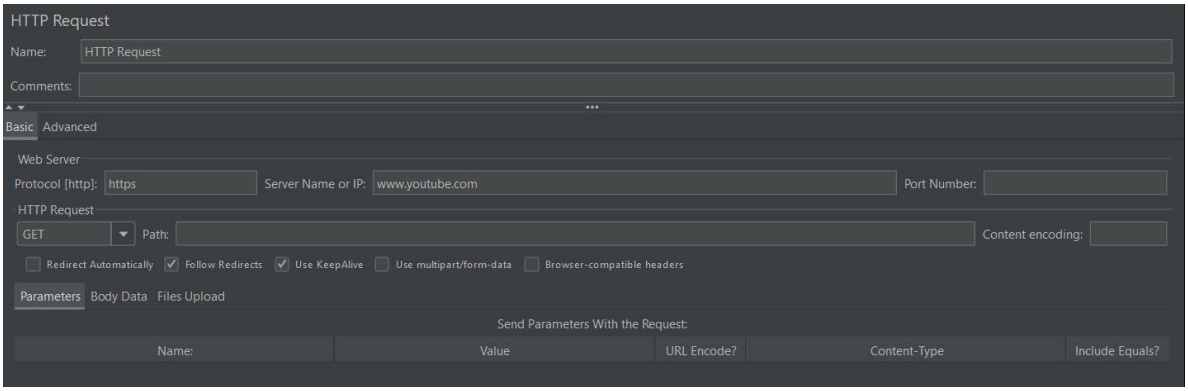
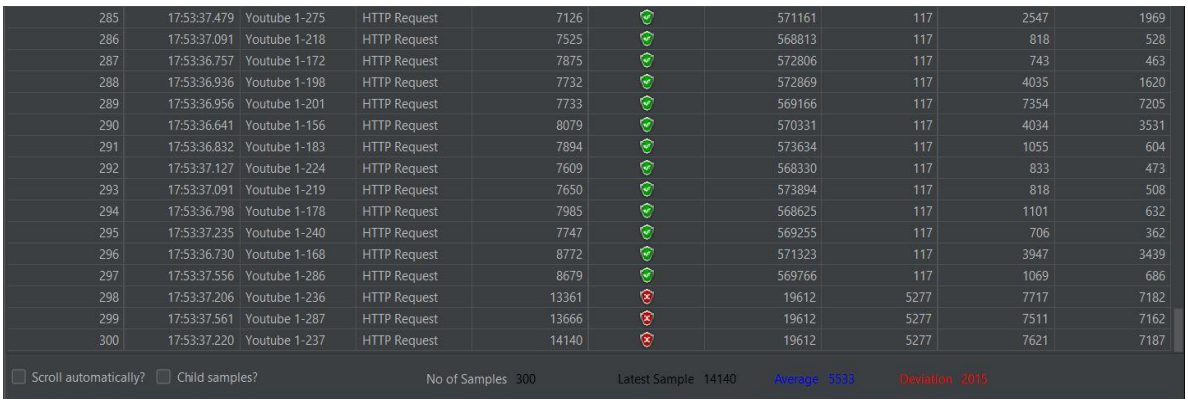


Figure: http request



285	17:53:37.479	Youtube 1-275	HTTP Request	7126		571161	117	2547	1969
286	17:53:37.091	Youtube 1-218	HTTP Request	7525		568813	117	818	528
287	17:53:36.757	Youtube 1-172	HTTP Request	7875		572806	117	743	463
288	17:53:36.936	Youtube 1-198	HTTP Request	7732		572869	117	4035	1620
289	17:53:36.956	Youtube 1-201	HTTP Request	7733		569166	117	7354	7205
290	17:53:36.641	Youtube 1-156	HTTP Request	8079		570331	117	4034	3531
291	17:53:36.832	Youtube 1-183	HTTP Request	7894		573634	117	1055	604
292	17:53:37.127	Youtube 1-224	HTTP Request	7609		568330	117	833	473
293	17:53:37.091	Youtube 1-219	HTTP Request	7650		573894	117	818	508
294	17:53:36.798	Youtube 1-178	HTTP Request	7985		568625	117	1101	632
295	17:53:37.235	Youtube 1-240	HTTP Request	7747		569255	117	706	362
296	17:53:36.730	Youtube 1-168	HTTP Request	8772		571323	117	3947	3439
297	17:53:37.556	Youtube 1-286	HTTP Request	8679		569766	117	1069	686
298	17:53:37.206	Youtube 1-236	HTTP Request	13361		19612	5277	7717	7182
299	17:53:37.561	Youtube 1-287	HTTP Request	13666		19612	5277	7511	7162
300	17:53:37.220	Youtube 1-237	HTTP Request	14140		19612	5277	7621	7187

Figure: view result table for 300 users

Scenario2:

Load testing: measure the YouTube performance to **the history page** under expected loads to ensure it can handle the anticipated user base without degrading performance

Thread properties: number of users: 300, ramp-up period (in sec): 2, loop count: 1

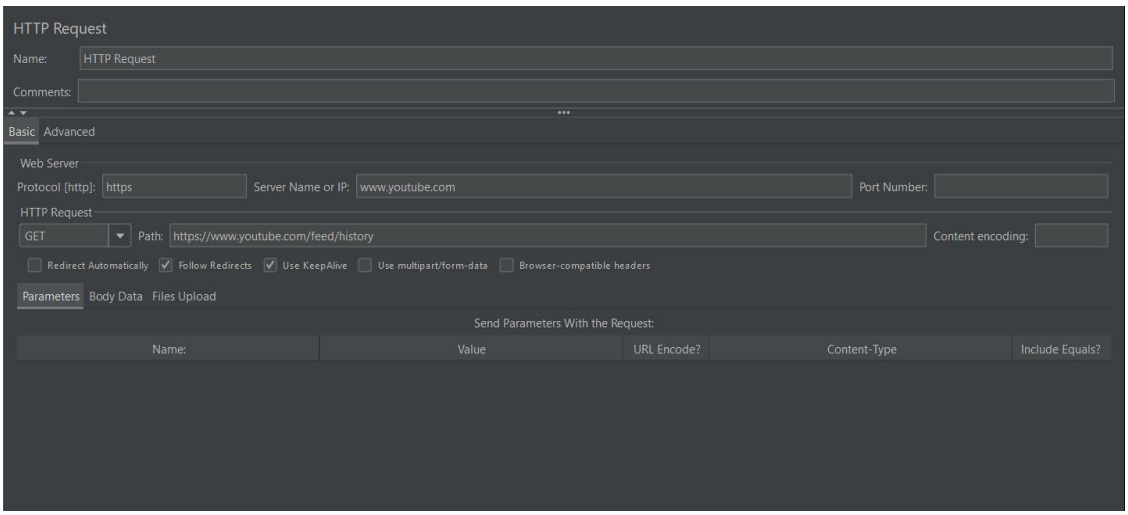
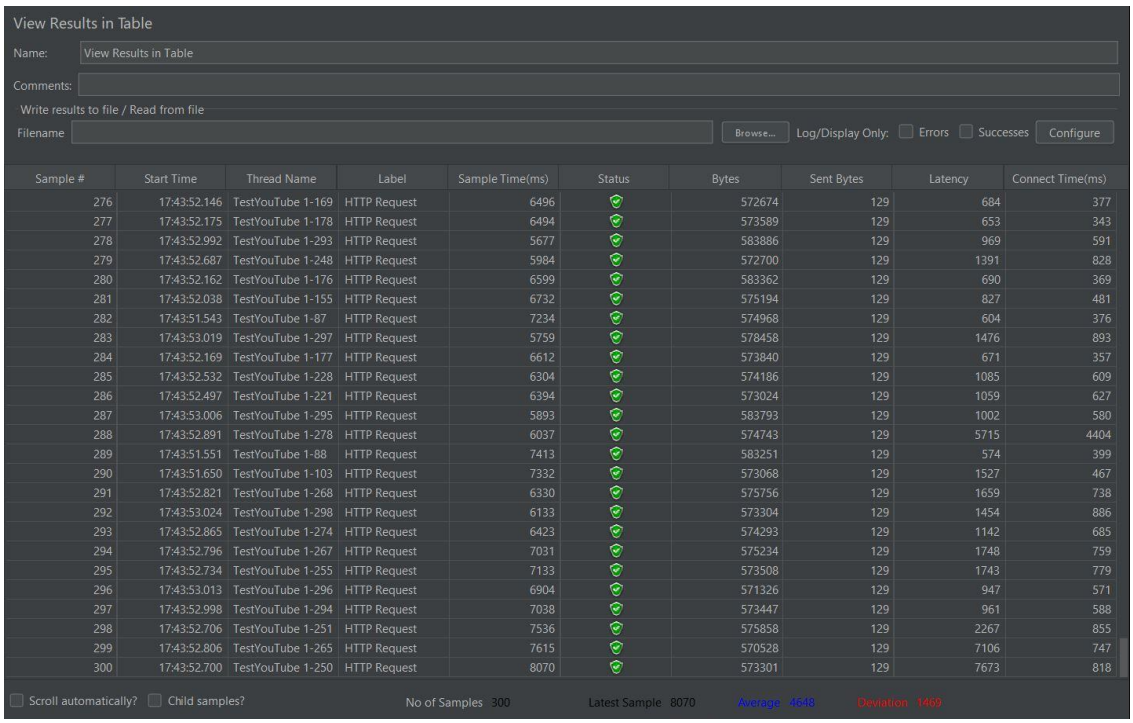


Figure: http request



Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
276	17:43:52.146	TestYouTube 1-169	HTTP Request	6496		572674	129	684	377
277	17:43:52.175	TestYouTube 1-178	HTTP Request	6494		573589	129	653	343
278	17:43:52.992	TestYouTube 1-293	HTTP Request	5677		583886	129	969	591
279	17:43:52.687	TestYouTube 1-248	HTTP Request	5984		572700	129	1391	828
280	17:43:52.162	TestYouTube 1-176	HTTP Request	6599		583362	129	690	369
281	17:43:52.038	TestYouTube 1-155	HTTP Request	6732		575194	129	827	481
282	17:43:51.543	TestYouTube 1-87	HTTP Request	7234		574968	129	604	376
283	17:43:53.019	TestYouTube 1-297	HTTP Request	5759		578458	129	1476	893
284	17:43:52.169	TestYouTube 1-177	HTTP Request	6612		573840	129	671	357
285	17:43:52.532	TestYouTube 1-228	HTTP Request	6304		574186	129	1085	609
286	17:43:52.497	TestYouTube 1-221	HTTP Request	6394		573024	129	1059	627
287	17:43:53.006	TestYouTube 1-295	HTTP Request	5893		583793	129	1002	580
288	17:43:52.891	TestYouTube 1-278	HTTP Request	6037		574743	129	5715	4404
289	17:43:51.551	TestYouTube 1-88	HTTP Request	7413		583251	129	574	399
290	17:43:51.650	TestYouTube 1-103	HTTP Request	7332		573068	129	1527	467
291	17:43:52.821	TestYouTube 1-268	HTTP Request	6330		575756	129	1659	738
292	17:43:53.024	TestYouTube 1-298	HTTP Request	6133		573304	129	1454	886
293	17:43:52.865	TestYouTube 1-274	HTTP Request	6423		574293	129	1142	685
294	17:43:52.796	TestYouTube 1-267	HTTP Request	7031		575234	129	1748	759
295	17:43:52.734	TestYouTube 1-255	HTTP Request	7133		573508	129	1743	779
296	17:43:53.013	TestYouTube 1-296	HTTP Request	6904		571326	129	947	571
297	17:43:52.998	TestYouTube 1-294	HTTP Request	7038		573447	129	961	588
298	17:43:52.706	TestYouTube 1-251	HTTP Request	7536		575858	129	2267	855
299	17:43:52.806	TestYouTube 1-265	HTTP Request	7615		570528	129	7106	747
300	17:43:52.700	TestYouTube 1-250	HTTP Request	8070		573301	129	7673	818

Figure: view result table for 300 users

Conclusion

This project focused on testing the YouTube website to ensure its functionality, performance, and reliability

By following a structured QA process, we evaluated the systems ability to handle different scenarios and provide a smooth user experience, the testing helped identify strengths and potential issues, ensuring that the website performs well under various conditions

Overall, this project highlighted the importance of thorough testing in improving software quality and ensuring that YouTube remains a reliable and user-friendly system